AI-Powered Visual Data Analytics and Business Intelligence Platform

Milestone 2 Report

Document Intelligence and Knowledge Base

PROJECT FOCUS

Searchable workplace safety knowledge from uploaded documents

MILESTONE 2 SCOPE

Document upload, text extraction, chunking, fact extraction, full-text search and source-based knowledge retrieval

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1. Project Idea

Milestone 2 extends the AI-Powered Visual Data Analytics platform with a document intelligence layer. While Milestone 1 focused on computer-vision analysis of workplace images and videos, this milestone enables users to upload workplace safety documents such as inspection reports, safety manuals and standard operating procedures, then extract, index and search the information contained in them.

The system extracts text from PDF, DOCX and TXT files, divides the content into searchable chunks, and applies rule-based extraction to identify structured safety information such as hazards, PPE requirements, corrective actions, dates, locations and recommendations. SQLite FTS5 provides full-text search across the uploaded document repository.

Search results and retrieved passages are linked to their source document and page reference wherever available. The Knowledge Base complements Milestone 1 by providing documentary context for workplace safety review. The Q&A interface retrieves supporting passages from uploaded documents; it does not generate answers using an LLM.

2. Milestone 2 Objectives

- Implement document upload and processing for PDF, DOCX and TXT files.
- Extract document text and preserve page-level source references wherever possible.
- Divide extracted text into overlapping chunks and index the text using SQLite FTS5.
- Apply rule-based extraction to identify hazards, PPE requirements, corrective actions, dates, locations and recommendations.
- Retrieve relevant passages using BM25-ranked full-text search with highlighted snippets.
- Provide a document question-and-answer interface that returns supporting passages from the indexed documents.
- Integrate the Knowledge Base with the existing PPE detection application without breaking Milestone 1 functionality.

3. Emerging Skills

- PDF text extraction using PyMuPDF and page-level references
- DOCX parsing using python-docx and approximate page estimation
- SQLite FTS5 full-text indexing, BM25 ranking and snippet generation
- Text chunking with overlap to preserve context between segments
- Rule-based information extraction using Python regular expressions
- REST API design for document upload, processing, retrieval and deletion
- React state management and upload progress handling
- Evidence-based search design and source traceability

4. Technology Stack

Component	Technology	Purpose
PDF extraction	PyMuPDF (pymupdf / fitz)	Extracts PDF text page by page and helps identify pages without extractable text.
DOCX extraction	python-docx	Reads Word document paragraphs and table cells; page numbering is approximate.
TXT extraction	Python built-in file handling	Reads plain-text files; page references are not available.
Full-text search	SQLite FTS5	Indexes chunk text and supports MATCH queries, BM25 ranking and highlighted snippets.
Structured extraction	Python re (regular expressions)	Identifies safety facts using rule-based patterns.
Backend	Flask + Flask-SQLAlchemy	Provides REST APIs for document processing, search and retrieval.
Relational storage	SQLite + SQLAlchemy	Stores document metadata, chunks and extracted facts.
Authentication	PyJWT + bcrypt	Provides JWT-based authentication and password hashing.
Frontend	React 18 + Vite	Builds the single-page application and Knowledge Base screens.
Routing / styling	React Router v6 + Tailwind CSS v4	Handles page navigation and interface styling.
HTTP / icons	Axios + lucide-react	Connects frontend to backend APIs and supplies interface icons.
Testing	pytest	Automated tests for extraction, search, facts, deletion and regression.

5. System Architecture

The milestone adds a document intelligence layer alongside the existing PPE detection pipeline. Both layers use the same Flask application, SQLite database and JWT authentication system.

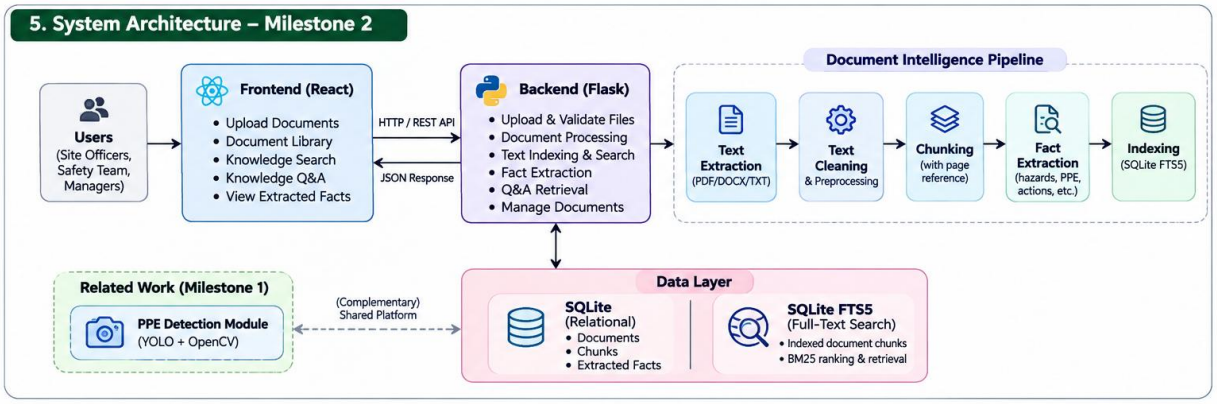


Figure 3. System architecture of the Milestone 2 document intelligence and knowledge-base module.

Layer	Main responsibility
React presentation layer	Document upload, Knowledge Base dashboard, document library, document reader and search screens.
Flask API layer	Validates uploads and exposes document, search and Knowledge Q&A endpoints.
Document processing	Extracts text from PDF, DOCX and TXT; divides text into 300-word chunks with a 50-word overlap.
Information extraction	Uses regular-expression rules to identify structured safety facts.
Search indexing	Stores chunk text in the SQLite FTS5 virtual table and ranks matching passages using BM25.
Data storage	Stores document metadata, chunks and extracted facts in SQLite; stores uploaded files in the documents/ directory.

Document processing workflow

React Upload UI → POST /api/documents (Flask) → file validation and storage → text extraction (PyMuPDF / python-docx / plain text) → 300-word chunking with 50-word overlap → regex-based fact extraction → SQLite FTS5 indexing → document status update → refreshed Document Library.

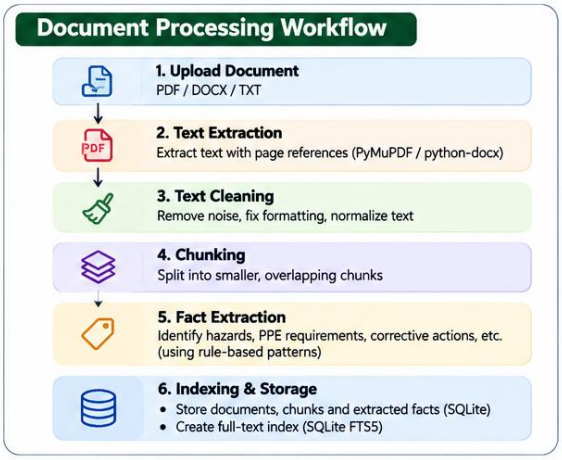


Figure 4. Document processing workflow from upload through extraction, chunking, fact extraction and indexing.

Search and retrieval workflow

React Search page → GET /api/search?q=... → Flask search route → FTS5 MATCH query and BM25 ranking → snippet generation → join with document and chunk records → JSON response containing document name, page and snippet → highlighted results displayed in React.

Knowledge Q&A workflow

React Q&A input → POST /api/knowledge/ask → retrieval from the same FTS5 index → top supporting passages returned with source document and page details. No LLM is used to generate an answer.

6. Project Working Diagram

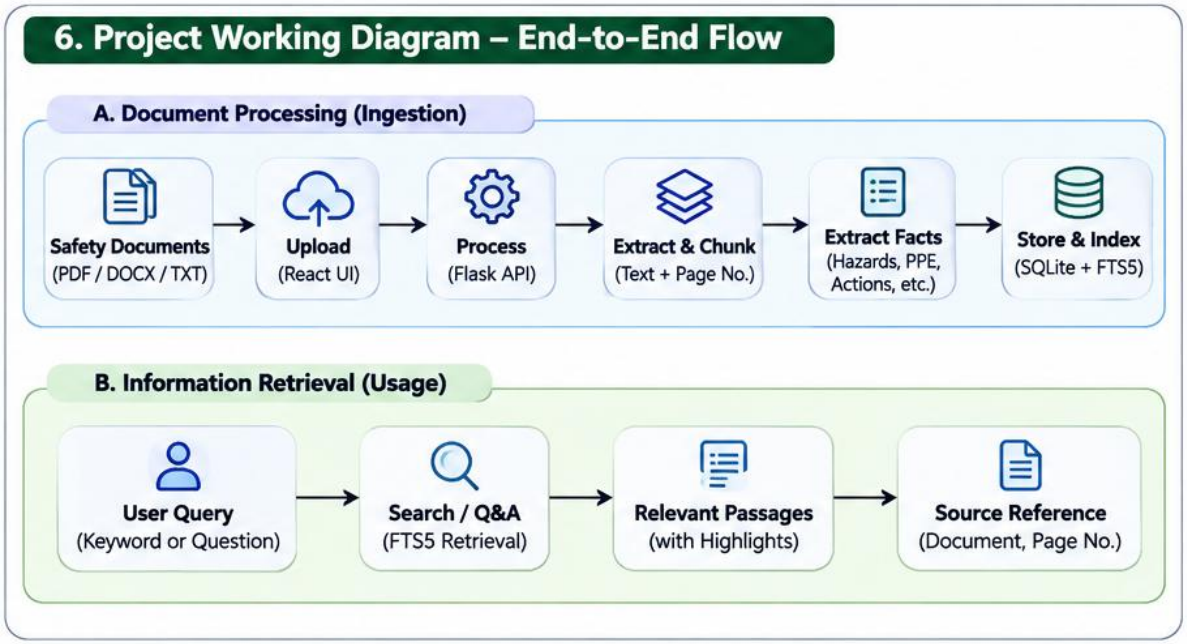


Figure 5. End-to-end project working diagram showing document ingestion and information retrieval.

Workflow	Sequence
Upload and indexing	React upload → Flask API → file storage → text extraction → chunking → rule-based fact extraction → FTS5 index → Document Library
Document search	Search query → Flask search API → FTS5 MATCH → BM25 ranking → highlighted snippets → source document and page reference
Knowledge Q&A	Question → retrieval from FTS5 → supporting passages → source links for verification

7. Class / Data Model

The following entities were added for Milestone 2. Milestone 1 entities (User, Analysis and Detection) remain unchanged.

Entity	Key fields	Relationship / purpose
Document	id, user_id, original_filename, stored_filename, file_type, file_size_bytes, status, chunk_count, fact_count, error_message, created_at, processed_at	Represents an uploaded safety document; has many DocumentChunks and ExtractedFacts.
DocumentChunk	id, document_id, text, chunk_index, page_number, word_count	Stores one text segment; linked to its source document and indexed through FTS5.
ExtractedFact	id, document_id, chunk_id, fact_type, value, source_text, page_number, confidence	Stores a rule-extracted fact with its source text and document reference.
document_fts	chunk_text, chunk_id, document_id, page_number	SQLite FTS5 virtual table containing searchable chunk text and source identifiers.

8. Milestone 2 Deliverables (Weeks 3–4)

- Document upload API supporting PDF, DOCX and TXT, with file-type validation and size checking.
- PDF text extraction using PyMuPDF, DOCX extraction using python-docx, and TXT reading using Python file handling.
- 300-word text chunks with 50-word overlap to retain context around chunk boundaries.
- Rule-based extraction of hazards, PPE requirements, corrective actions, dates and locations.
- SQLite FTS5 full-text index with BM25 ranking, porter stemming and snippet generation.

- Document and search REST API routes for upload, listing, statistics, detail, chunks, facts, reprocessing, deletion, download, search and Knowledge Q&A.
- Knowledge Base Dashboard with document, index, chunk and fact statistics.
- Document Library with drag-and-drop upload, progress feedback, status/type filters and document actions.
- Document Reader with document metadata, extracted-fact groups and paginated text chunks.
- Search page with highlighted matching snippets, file-type filters, URL-synchronized query state and Knowledge Q&A.

9. Application Screens and Demonstration

The following screenshots were captured from the running SafeVision AI application. They show the implemented Knowledge Base dashboard, document library, document reader, document search and Knowledge Q&A interface.

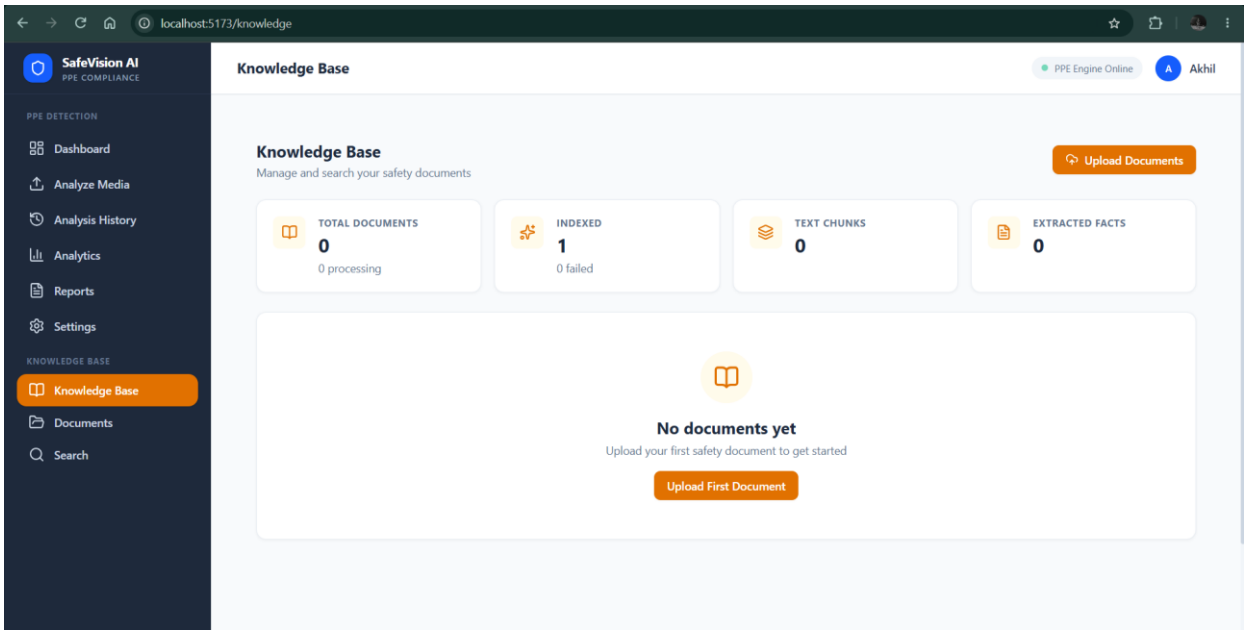


Figure 1. Knowledge Base dashboard showing document/index summary cards and the upload entry point.

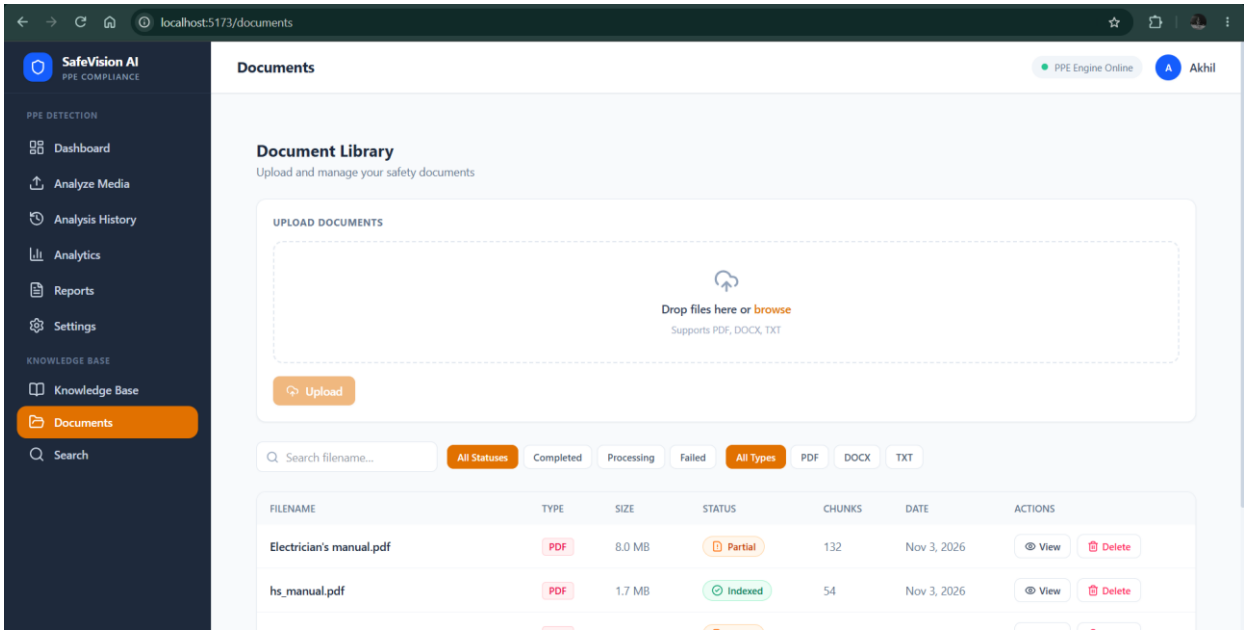


Figure 2. Document Library upload area and document list with file type, size, processing status, chunk count and actions.

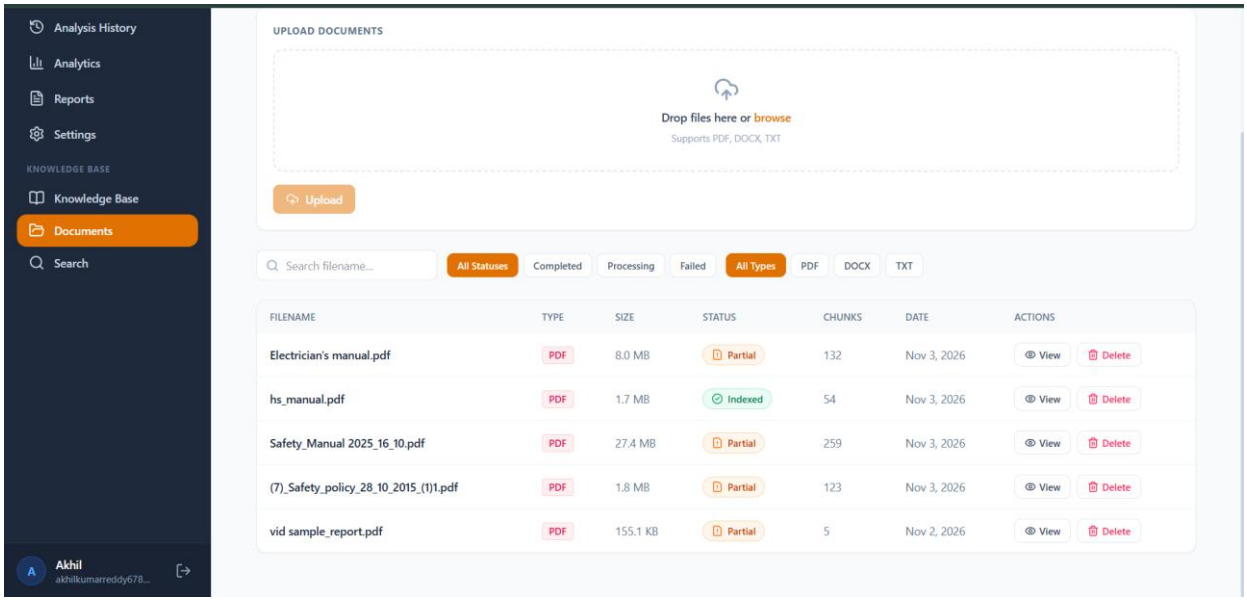


Figure 3. Document Library showing uploaded safety manuals and their Indexed or Partial processing status.

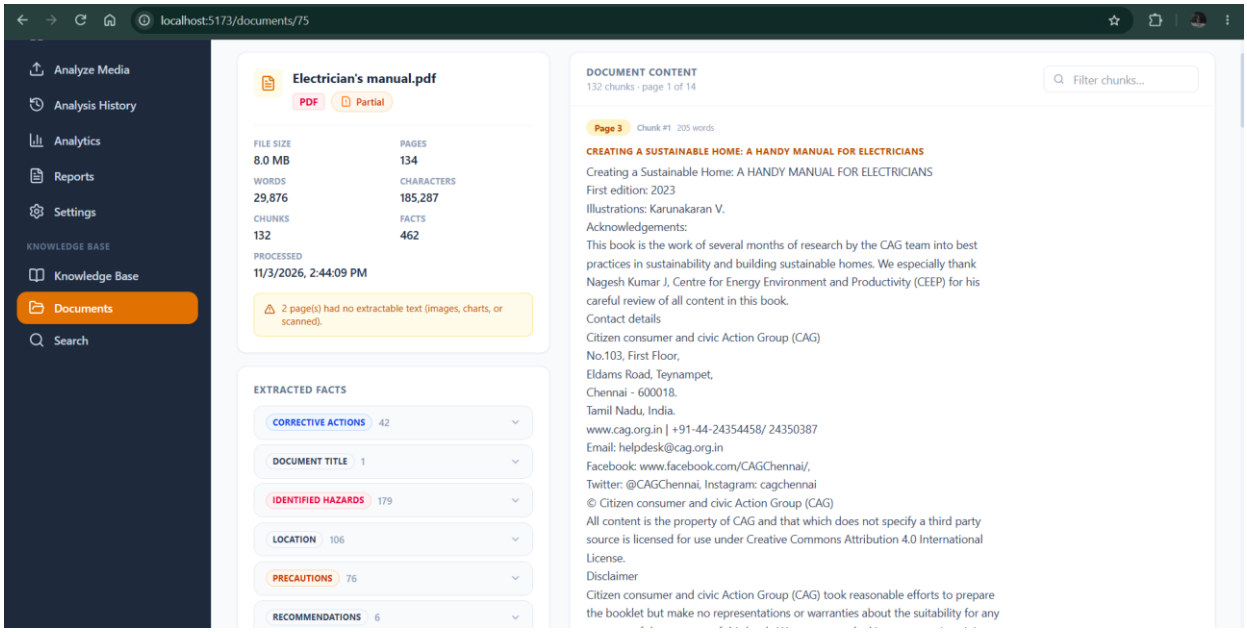


Figure 4. Document Reader showing document metadata, extracted-fact categories and paginated text chunks.

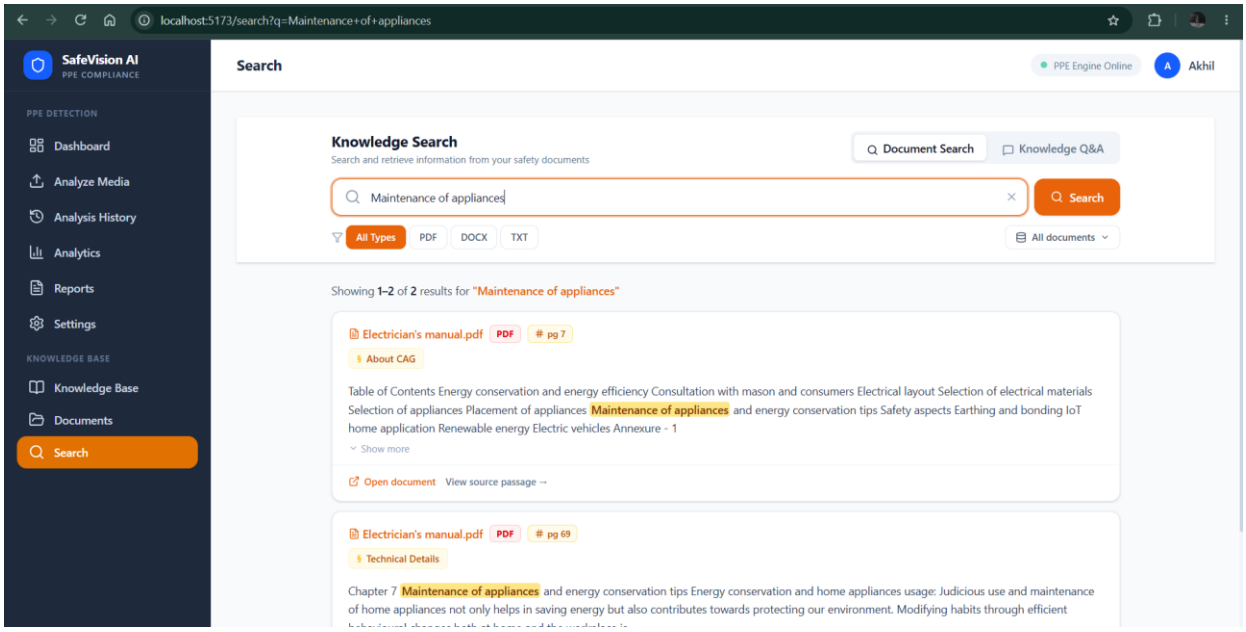


Figure 5. Document Search results showing matching text snippets, highlighted query terms, source filename and page references.

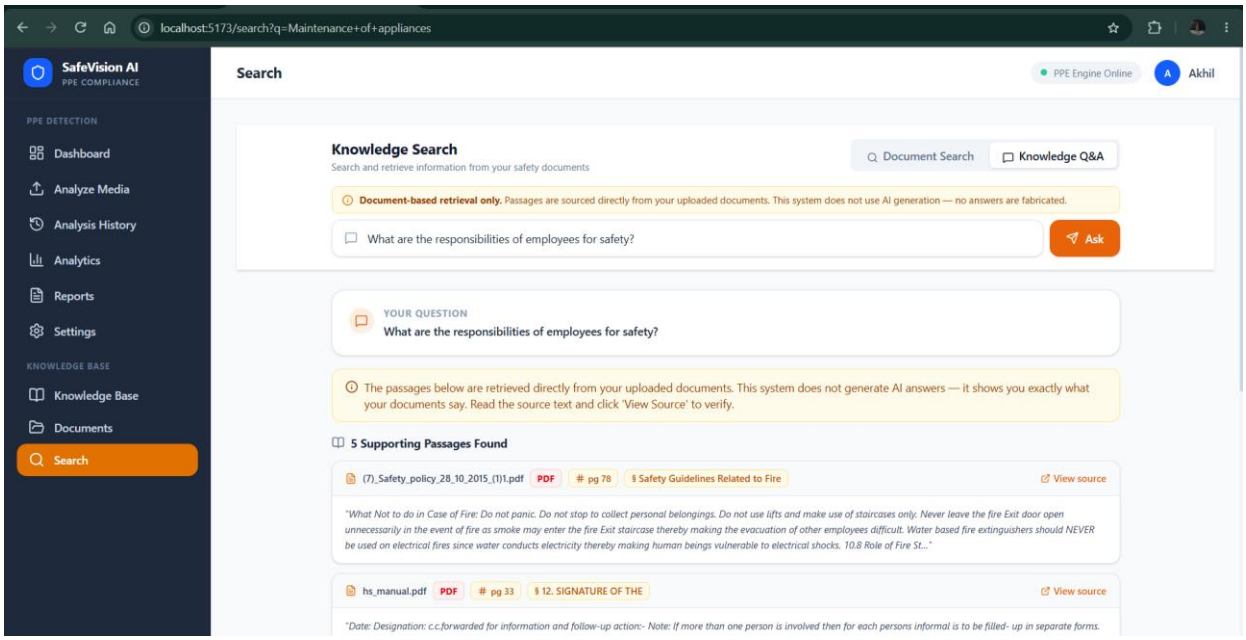


Figure 6. Knowledge Q&A interface showing the submitted question and supporting passages retrieved from uploaded documents.

10. Validation and Test Results

The milestone report records 27 automated tests passing with 0 failures. The test coverage is summarized below.

Test area	Coverage	Result
File validation	Unsupported types, empty/missing files and unauthenticated requests	PASSED (4/4)
Text extraction	TXT and DOCX extraction, text accuracy and corrupted-file handling	PASSED (5/5)
Page references	TXT page references and approximate DOCX page references	PASSED (2/2)
Search	Known terms, matching snippets, irrelevant terms, empty queries and special characters	PASSED (5/5)
Information extraction	Extracted facts and source-text references	PASSED (2/2)
Reprocessing	Prevents duplicate chunks when a document is reprocessed	PASSED (1/1)
Deletion	Removes document records and corresponding FTS search entries	PASSED (2/2)
Document API	Statistics, document listing and Knowledge Q&A retrieval behavior	PASSED (4/4)
Milestone 1 regression	PPE analysis API and dashboard statistics remain accessible	PASSED (2/2)

Total recorded result: 27 passed, 0 failed. The reported test duration was approximately 2 minutes 13 seconds, including model loading for regression tests.

11. Challenges and Solutions

Challenge	Cause	Solution applied
FTS5 error: no such column: T.chunk_text	The external-content FTS table column did not match the source table column name.	Removed the content= option; FTS5 stores chunk text directly in its virtual table.
FTS5 limitation with in-memory test database	A separate SQLite connection could not access the in-memory FTS virtual table created by another connection.	Added configure() logic to detect in-memory SQLite and disable FTS indexing for those tests.
PyMuPDF deprecation warning	The newer PyMuPDF API recommends importing pymupdf rather than fitz.	Recorded as a non-breaking warning; current extraction still works, with an update noted for later.
Video codec compatibility (Milestone 1 carry-over)	The avc1 codec is not available in every Windows OpenCV build.	Used mp4v as a fallback; ffmpeg re-encoding can be used if required.
Large JavaScript bundle warning	Frontend dependencies increase the Vite bundle size.	Treated as non-blocking; the report records a gzipped bundle size of approximately 244 KB.

12. Conclusion

Milestone 2 adds a document intelligence and knowledge base layer to the existing PPE detection platform. The application accepts PDF, DOCX and TXT safety documents, extracts their text, divides the content into searchable chunks, identifies structured safety facts using rule-based patterns and indexes the text for retrieval.

The Search page retrieves matching passages using SQLite FTS5 and BM25 ranking, while the Knowledge Q&A interface returns supporting passages from the same document index. Source document names and page references help users verify the retrieved information. The milestone report records 27 automated tests passing, including a regression check for Milestone 1 functionality.

Together, Milestones 1 and 2 provide two complementary sources of workplace safety information: visual PPE detection from images and videos, and searchable documentary evidence from safety manuals and reports.